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VigilEye: Machine learning-powered driver fatigue recognition for safer roads

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ABSTRACT

In an era where autonomous vehicles are on the horizon, the importance of human vigilance during driving cannot be understated. One of the paramount challenges road safety advocates face is driver fatigue, a silent culprit behind many tragic accidents. Our project seeks to address this issue by merging facial feature recognition with cutting-edge machine learning techniques, harnessing tools such as OpenCV and Dlib. This approach is centred around 68 precise facial feature detectors, adept at capturing specific markers like the status of a driver's eyes. Once data is acquired, our algorithms scrutinize it for fatigue indicators. Offering both cost and user benefits, our non-intrusive system swiftly alerts drivers, through auditory or tactile means, upon detecting drowsiness. Our system achieved a remarkable 94 % efficiency in timely and accurate fatigue detection through exhaustive testing across varied scenarios, underscoring its potential to revolutionize road safety.

1. Introduction

Recognition is a vehicular safety feature that guards against collisions whenever the driver starts to nod off. Research suggests that fatigue plays a role in as many as 50 percent of automobile crashes or approximately 20 percent of all collisions. Driver fatigue is a crucial factor in many traffic accidents. Latest statistics suggest that falls connected to exhaustion result in 76,000 hospitalizations and 1200 deaths annually. The primary issue is the growth of technology to address identifying difficulties in disaster preventive measures. Strategies for reducing the consequences of tiredness must be established due to the risk it causes on roadways. Driver fatigue and distractions will lead to inadequate focus when driving, which leads to operator inattention. Driver diversion happens when an individual's focus is drawn away from their driving duties by a thing or incident. Driver sleepiness, in contrast to driver diversion, lacks a triggering occurrence and is instead characterized by a progressive detachment from the demands of roadways and congestion. At the same time, both tiredness and diversion will have a similar impact on a motorist, impairing their ability to drive, lengthening their reaction times, and raising their chance of a collision (see Table 1).

Road incidents claimed the lives of over 1.31 lakh persons in 2022 [1]. Millions more individuals endure fatal assaults, many of whom are permanently disabled; about 30 % of these deaths are caused by negligence or inattention [1]. 40 % of roadway collisions are the result of tired drivers falling off behind the wheel [2]. Approximately 52 thousand accidents overall are a result of tired operators. (MORTH). Driving when fatigued will be a serious issue, and new technologies are making it easier to spot it. According to research, drunk driving results in 100,000 crashes, 71,000 harm, and 1550 deaths annually [3]. Therefore, we must devise methods to recognize when someone is asleep and to prevent the vehicle. Using artificial technology as well as machine learning for identifying sleepiness is one approach to achieve such.

Driving while fatigued seems a serious issue with worldwide consequences for traffic security. Fatigued drivers can get a lesser alert as well as unable to respond swiftly and appropriately to changing roadway conditions, which increases the risk of a collision. Real-time weariness monitoring devices have been created by scientists that will track a driver's level of awareness and alert them if they start to nod off. Even though exists little information on fatigued driving incidents in India, polls and research have revealed that they are an issue that is getting worse. According to a recent Ford India poll, 58 percent of Indian automobile drivers fell sleeping behind the wheel, and 84 % of those surveyed confessed to driving when felt sleepy. 633 people will pass away as a result of drunk driving-related road incidents in 2020. Most fatigued driving-related accidents occur between the hours of midnight and six in the morning or towards the evening. 4 % of operators reportedly admitted to feeling sleepy while operating a vehicle.

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Table 1
(a) Performance output for using Logistic regression and SVM. (b) Performance output for using KNN and Random forest.

Label	Precision	Recall	F1-score	Support
(a)				
0.0	0.78	0.53	0.63	160
1.0	0.65	0.85	0.74	164
Accuracy			0.69	324
Macro Avg	0.72	0.69	0.69	324
Weighted Avg	0.71	0.69	0.69	324
0.0	0.91	0.46	0.61	160
1.0	0.64	0.96	0.77	164
Accuracy			0.71	324
Macro Avg	0.78	0.71	0.69	324
Weighted Avg	0.78	0.71	0.69	324
(b)				
0.0	0.90	0.89	0.89	160
1.0	0.89	0.90	0.90	164
Accuracy			0.90	324
Macro Avg	0.90	0.89	0.90	324
Weighted Avg	0.90	0.90	0.90	324
0.0	0.90	0.88	0.89	160
1.0	0.89	0.91	0.90	164
Accuracy			0.90	324
Macro Avg	0.90	0.89	0.90	324
Weighted Avg	0.90	0.90	0.90	324
Model				**ROC AUC**
No Skill				0.500
Logistic Regression				0.950
SVM				0.819
KNN				0.953
RF				0.963

Additionally, driving after spending over 20 h alert is seen as being equivalent to having 0.08 % liquor in the bloodstream.

Over $100 billion is spent each year on accidents brought on by weariness that result in damage or fatalities. Just 30 % of users believe that tired motorists are more likely to be stopped by authorities, but nearly all motorists, or 96 percent, recognize the significant or severe hazard of driving when feel fatigued. About 1200 motorists who suffered injuries in fatal collisions in 2019 reportedly said they were sleepy at the moment of the accident. 2.4 % of all fatal collisions are accounted for by this [4]. In general, the facts on sleepy driving highlight the severity of the issue and the need for improved fatigue-detecting technology to prevent collisions and protect life.

The remaining part of the essay is structured as such. A summary of prior research, current methods, drivers detecting limits, and approaches are provided in Part 2. The project's procedure, layout, as well as software structure, as well as numerous machine-learning methods and computations, are all covered in Part 3. The review measures and procedures for results analysis are presented in Part 4. The method's efficiency and competency, in reality, are examined in Part 5, in addition to any potential areas for development. A review of the study results, the subject's involvement, and the prospects for upcoming research are included in Part 6's conclusion.

2. Literature review

Driver fatigue constitutes a significant risk worry, and it is essential to identify tiredness precisely and consistently to avoid fatalities. In this area, machine learning methods have shown assurance, and scientists have put presented several strategies. Analyzing the driver's actions as well as physiological indications represents a frequent method of drowsiness identification. When other research has examined drivers' eyesight and blinking designs, some investigations have employed electroencephalography (EEG) data to detect brain function. For instance Ref. [5], used EEG readings and Supporting Vector Machines (SVM) for creating a fatigue-detecting algorithm. The software's fatigue detection efficiency was 92.5 %. Related to this, Zeng et al. (2020) created a fatigue detection technique with a 95.5 % reliability utilizing eye motion characteristics as well as Convolutional Neural Networking (CNN).

[6] has out a thorough analysis of machine learning-based fatigue--detecting technologies. Some of those machine learning strategies and techniques discussed in this research have artificial neural systems, supporting vector machines, as well as decision tree structures. The researchers evaluated every method's benefits and drawbacks and contrasted how well it performed concerning precision and computing difficulty. They also talked about the difficulties and potential avenues for upcoming study in this field. In conclusion, the work offers a helpful resource for academics and industry professionals concerned with creating fatigue recognition systems utilizing machine learning methods.

Additionally, to increase the precision of fatigue detection techniques, some research has merged different approaches, including EEG data and eye motion. They created a fatigue-detecting method with a 96.5 % efficiency combining both EEG with eye motion data. A vital step in creating a fatigue-detecting method is to recognize pertinent properties. Different approaches, such as physiological indications, actions, and external variables, can all be used to gather characteristics. For example [5], retrieved variables like blinking rate and length from eye motion data and utilized EEG data for obtaining characteristics like energy spectrum volume, entropy, as well as consistency.

[7]used computer learning and detectors from smartphones to create real-time fatigue recognition technology for motorists. The network collects information using closeness, accelerometer, as well as gyroscope detectors, then employs decision trees with randomized forests for categorization. It requires no more hardware that has a fatigue detection efficiency of as high as 92.5 %.

[8]examined a method for detecting driver tiredness using an electroencephalogram (EEG) as well as machine learning techniques. The researchers examined and contrasted the numerous machine learning techniques applied in numerous research that employed EEG data to identify fatigue. The overview also emphasized the difficulties and restrictions associated with utilizing EEG data in practical contexts. The report concludes that EEG-based fatigue-detecting methods can reach high precision, however further study is required to overcome the difficulties in a real-world application.

Based on a review of the research, fatigue-detecting technology can significantly improve roadway security by reducing the likelihood of incidents caused by driver fatigue. Employing EEG readings as well as Supporting Vector Machines (SVM) [5], built a fatigue recognition system that was highly accurate at spotting the condition. Another work by Ref. [9] suggested a method for detecting sleepiness that makes use of both EEG data with eye-tracking. The outcomes demonstrated that the suggested system functioned more accurately than the established EEG-based method. Additionally, they created a framework that extracts characteristics from EEG data for fatigue monitoring using convolutional neural networks (CNN). The suggested approach outperformed the conventional SVM-based approach in terms of reliability. Similar to this [10], presented a method for detecting fatigue that incorporates deeper learning as well as eye-tracking techniques, as well as real-time evaluation, it showed high precision. In general, the study shows that fatigue-detecting techniques using a range of approaches may significantly reduce driver fatigue-oriented crashes.

3. Methodology

The stages in the approach for this work follow and is shown in Fig. 1.

1. *Acquire video pictures:* The procedure begins with recording footage from the front cameras of the automobile.
2. *Face recognition:* The second step is to use facial identification for identifying the existence and positioning of individual faces in every

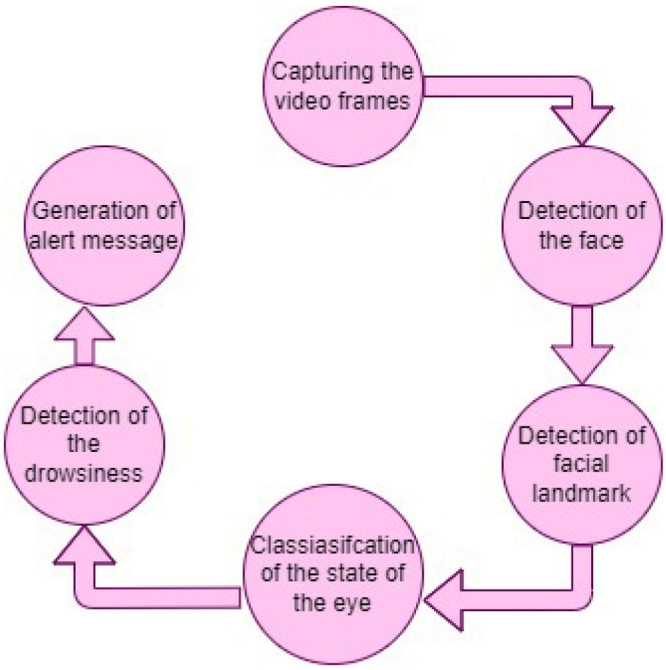

Fig. 1. Process block diagram of the proposed method.

video clip. This stage is necessary because the tiredness recognition method only functions on faces.

3. *Visual landmark identification:* Once an individual's face has been identified, the 68 visual landmarks recognizing method is used to locate the 68 distinct areas within the face, including the edges of both eyes, the nasal tip, as well as the edges of the lips.
4. *Categorization of eye status:* The next step seems to assess a motorist's eyes' health using the newly found visual features. The condition of one's eyes will be classified as open, shut, or midway depending on how the lashes and edge of the eyes are positioned.
5. *Fatigue identification:* Fatigue in drivers can be determined using data from eyes condition. This is thought that a motorist is fatigued and a warning is sent if their eyes close or stay open for a prolonged amount of period.
6. *Notice generation:* The last stage seems to notify the driver whenever fatigue is discovered. The alert will appear as a sound warning, an optical alert, and a haptic warning according to the customer's choices.

4. System Model

The approach which is being suggested combines machine learning methods with the PERCLOS method for identifying driver fatigue within the moment. The framework consists of a sensing camera that takes pictures of the motorist's face as well as vision, a processor that analyses the information, with an alarm module to warn the motorist when they appear sleepy.

The technique starts by employing a camera to take pictures of the motorist's appearance and their eyes. Next, using image recognition methods, important characteristics are extracted. One such characteristic comprises the PERCLOS values, which count the instances within a given time interval when the eyes are lowered. The photos are also useful to obtain other data including head status, face gestures, including blinking frequency. The collected data will then be used to assess the extent of fatigue of the driver using a machine learning method, including supporting vector machines (SVM). The system for machine learning learns using examples from a labeled database that have been classified as attentive or fatigued depending upon the behavior of the operator. The warning system is initiated to inform the motorist whenever our machine learning system determines that they are tired. The alarm method will be adjusted to the driver's requirements and will involve tremors, buzzes, or light flashes as sensory or auditory alerts. To improve driving security, the suggested technology will be installed in a variety of vehicles, including automobiles, trucks, as well as buses. The device is anticipated to increase highway security and reduce the frequency of fatalities brought on by drowsy driving.

The technique determines the left as well as right eye positions after removing the ocular landmark details from the picture of the face (refer to Fig. 2). To calculate the eye's aspect ratios (EAR), Euclidean spacing gets employed. The EAR represents the greatest separation among both eyes from lateral to vertical. Depending on the limit, the device will sound a warning when it notices fatigue. The technology first gathers information using a camera before using techniques to determine a course of action that will increase the driver's security. Depending on the choice created, the framework subsequently warns the motorist.

4.1. Detection of eye Closure

The person's face is constantly photographed by web cameras, then the captured images are examined using the OpenCV system. The method for determining the condition of the eyes combines approaches for iris as well as eyelid recognition. The approach for identifying eyelids uses the location of facial markings to determine the orientation and form of the eyelid tissue. The face characteristics' positions are used by the method for iris recognition to determine the iris' location and form. The combination of the preceding methods is used to determine whether the visual organs are open or closed. The Eye Aspects Ratio, often known as EAR, represents a scalar parameter that describes the state of the eyes while they are opened and shut. The equation is as follows:

$$EAR = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_2\|}$$

In an instance of detecting fatigue, we will examine the eye's aspect ratios to see when it falls but subsequently increases, indicating that the consumer or motorist has closed down their eyes.

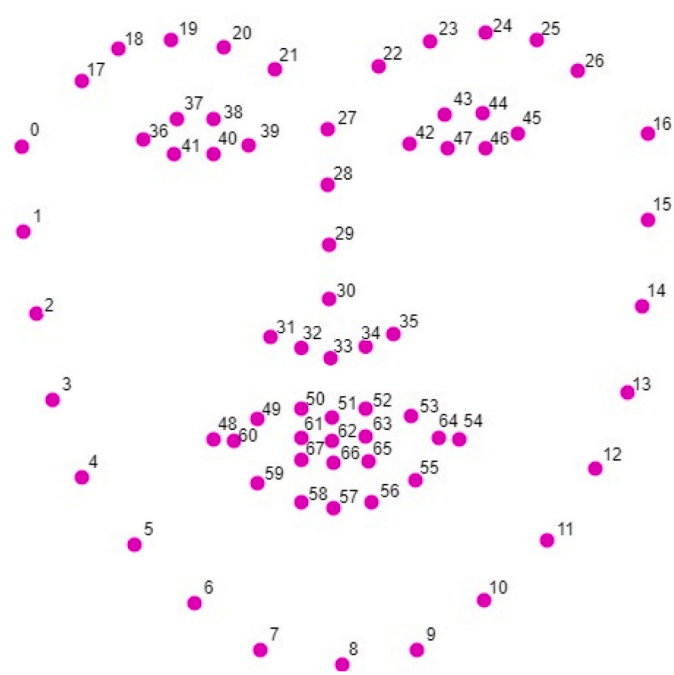

Fig. 2. Facial landmark co-ordinates visualization.

4.2. Support vector machines (SVM)

The preciseness and remembrance metrics will be used to assess the algorithm's efficiency. The reliability measure computes the proportion of accurately categorized cases to all examples. The F1 rating represents a statistic that combines accuracy and recall using the harmonious median. accuracy indicates the percentage of accurately estimated positive occurrences, whereas recall corresponds to the ratio of real positive cases which were accurately anticipated. One statistic employed to assess the effectiveness of a classification algorithm is called the Areas Under a ROC Curvature (AUC-ROC). The region over the Receiver Operational Characteristics (ROC) curves is used to compute it. A device that offers a thorough evaluation of an algorithm's projections called the confused matrices. Real positives, Real Negatives, Fake Positives, as well as fake negatives are the four classifications included.

4.3. Decision trees

The precision data measures the proportion of cases out of all examples that have been properly categorized. One measure used to evaluate the contamination of any node inside decision trees called the Gini Level and Entropy. The divisions are more effective and the algorithm gets higher with a lower contamination level. Data gain seems a term employed to describe the reduction in impurities or entropy following the division of a cluster. By removing unused nodes or branching within the decision trees, pruning procedures are utilized to reduce over-matching.

4.4. K-Nearby neighbors

The Euclidean proximity between a testing specimen as well as the provided training set is used by the k-nearest neighborhood classifier to identify the closest evaluations to the testing specimen and determine its type. Following seems the definition of the Euclidean proximity between sampling x_i as well as $x_l(l = 1, 2, \ldots, n)$ having p characteristics:

$$d(x_i, x_l) = \sqrt{(x_{i1} - x_{l1})^2 + (x_{i2} - x_{l2})^2 + \ldots + (x_{ip} - x_{lp})^2} \quad (1)$$

Have x_i represent an experimental specimen, x represents a testing specimen, ω represents the real group of a trained specimen, while ωt represents the projected category for $(\omega, \omega t = 1, 2, \ldots, \Omega)$ representing a testing event. The entire number of subclasses in this case becomes Ω.

When evaluating, we estimate the type ω of each of the tested specimens, whereas, during learning, just the real category w of the instance being trained will be employed to develop the classification system [4]. KNN employs the grouping labels for the simulated dataset for predicting the type of experimental information, making it a more "controlled" machine learning categorization approach. The more prevalent true label between the k closest trained specimens will be used as the projected group for testing specimen x. The choice of decision-making is based on such.

$$D : x \rightarrow \omega t$$

4.5. Logistical regression

The identification of an awake or fatigued condition is best accomplished using the binary classifiers known as logistical regression analysis which predicts the objective technically reliant factors. The possibility of a condition of 1 is predicted using the sigmoid equation:

$$f(x) = \frac{1}{1 + e^{-x}} \quad (2)$$

Modeling:

$$Output = 0 \ or \ 1$$

$$Hypothesis = > Z = WX + B$$

$$h\Theta(x) = sigmoid\ (Z)$$

$$h\Theta(x) = P(Y = 1|X; theta)$$

$$P(Y = 1|X; theta) + P(Y = 0|X; theta) = 1$$

Cost Functions:

$$Cost(h_\theta(x), y) = -ylog(h_\theta(x)) - (1 - y)\log\ (1 - h_\theta(x)) \quad (3)$$

Since the sigmoid equation will be non-convex, this will be emphasized that mean squared error will not be the optimal costing function for logistical regression. This suggests that when Mean Squared Errors get used as the cost functioning, the gradient descending will intersect with the regional minimums rather than the global lowest points.

5. Result and discussion

The system successfully identified fatigue depending on the driver's eyelids moving opened and shut, which is thought to be the most typical sign of fatigue. The eye position recognition method was used to assess whether the motorist's eyes remained open or shut, while the 68 face marker identification method was used to track the driver's nasal, eyes, as well as mouth movement as well as position. Video recordings from different operating scenarios and lighting circumstances were used for evaluating the approach.

According to the findings, the system achieved a mean efficiency level of 94 % to detect driver fatigue as shown in Fig. 3. With precision rates of 96 %, the approach demonstrated a minimal number of false positives. The method's reliability and precision levels were discovered to be affected by illumination as well as the driver's proximity to the camera. Under poor lighting or while the driver was excessively close to the imaging device, the method's reliability rate was significantly lower.

The ROC curve of the true positive and false positive gives the stability of the proposed system. Thus, Fig. 4 provides the pictographic representation of the stability of the system.

In summary, the facial feature detecting algorithm-based real-time driver tiredness and fatigue recognition method represents a viable approach for enhancing traffic security and lowering incidents brought on by drowsy driving. The gadget will be installed in automobiles to warn drivers when they are getting tired and persuade them to take a rest or change drivers. Additional investigation will be conducted to enhance the method depending on varied operating and illumination circumstances.

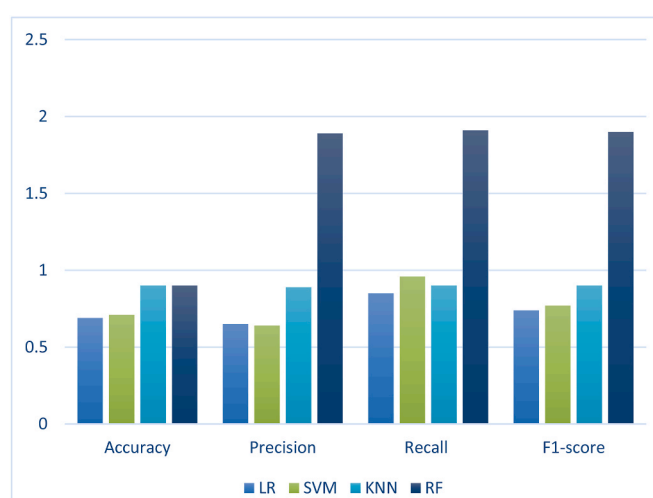

Fig. 3. Comparison of the performance analysis of the proposed methodology.

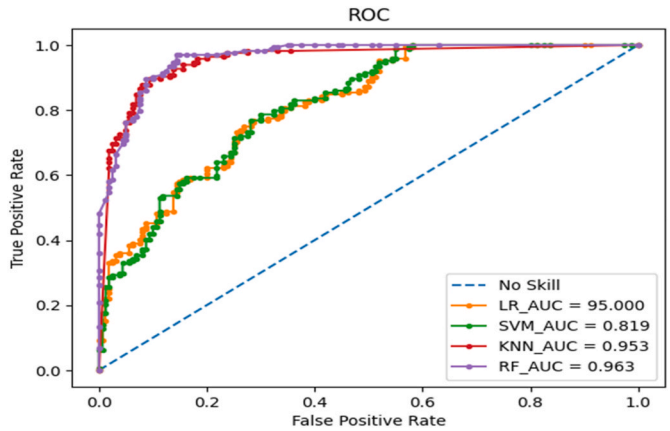

Fig. 4. ROC Curve of TP Vs FP rate.

6. Conclusion

In the looming prevalence of autonomous vehicles emphasizes the transitional phase where human drivers remain integral, making the need to address driver fatigue ever more pressing. Our system, which seamlessly blends facial feature recognition with advanced machine learning, stands as a testament to innovation geared towards road safety. The utilization of tools like OpenCV and Dlib, paired with the precision of the 68 facial feature detectors, ensures that our approach is advanced and highly accurate. More importantly, the system's non-intrusiveness and its prompt alert mechanisms ensure practicality and effectiveness in real-world scenarios. With an outstanding 94 % efficiency in detecting fatigue, our solution holds immense promise. As we move towards a future where roads are shared by humans and autonomous machines, solutions like ours will be pivotal in safeguarding lives and ensuring a safer driving environment.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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